Assignment 2 - Git & GitHub Fundamentals Report

Group Size: 2 members
Chosen Theme: Unit Converter
Full Names: Hafsa Borouja / Zulaikha Humayun
Student IDs: 145625 / 195640
Course Codes & Sections: CSC3324 02
GitHub Usernames: hafsabor1 / zulaikhahumayun21
GitHub Repository URL: https://github.com/zulaikhahumayun21/unit-converter
GitHub Pages URL: https://zulaikhahumayun21.github.io/unit-converter/

1. Project & Role Distribution

The group chose the Unit Converter theme and followed the 3-member path. The application is organized into separate Python modules for length, temperature and weight conversion, with a command-line entry point, pytest tests, requirements.txt and a static website under docs/. Responsibilities: Zulaikha Humayun (@zulaikhahumayun21) - length; Hafsa Borouja (@hafsabor1) - temperature.

2. Branches, Commits & Contributions

Hafsa - feature/temperature: temperature conversion (C, F, K), input validation, expanded tests, and the CI workflow with website asset organization. Commits: temperature input validation; expanded temperature tests; "ci: add automated tests workflow and organize website assets" (1815dc6).

Zulaikha – length feature: length conversion module and tests.

3. Pull Requests & Code Review

PR #1 (Feature/temperature) was authored by Hafsa and reviewed by Zulaikha.

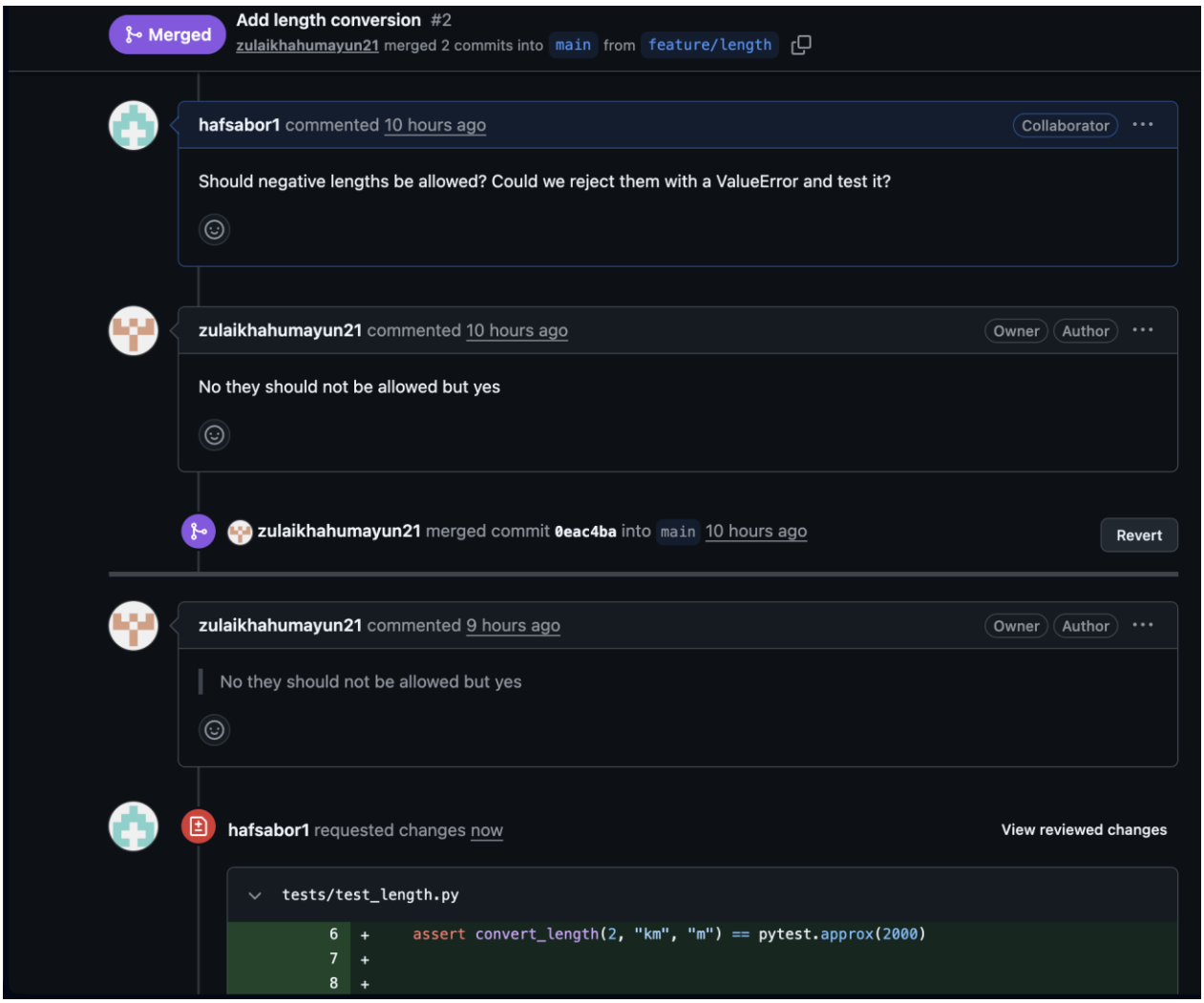
PR #2 (Add length conversion) was authored by Zulaikha and reviewed by Hafsa.

Request changes cycle: the reviewer left a line comment on Files changed and submitted Request changes . The author fixed it on the same branch, committed and pushed; the existing PR was reviewed again and approved before merging.

Request Changes and Correction Cycle

During the review of the length conversion feature, the reviewer left a line comment requesting an additional test for negative length inputs. This test helps verify that invalid negative values raise a ValueError.

The author was asked to add the test to tests/test_length.py, commit the correction, and push it to the same feature branch used by the existing pull request. The updated pull request should then be reviewed again, and approved once the requested correction is verified.



4. Merge Conflict

To practice resolving merge conflicts, two team members created separate branches from the same version of main and modified the same line in README.md:

Project status: Development in progress.

The first member's change was merged into main through a Pull Request. The second member then updated their branch using:

git fetch origin
git merge origin/main

Git detected a merge conflict because both branches had modified the same line differently. The **Current Change** represented the version from the second member's branch, while the **Incoming Change** represented the version already merged into main. After resolving the conflict, the file was staged, committed, and pushed to the second branch. The updated Pull Request was then reviewed by the third team member.

5. Unit Tests

The repository contains separate test files for the three conversion features: tests/test_length.py, tests/test_temperature.py, and tests/test_weight.py.

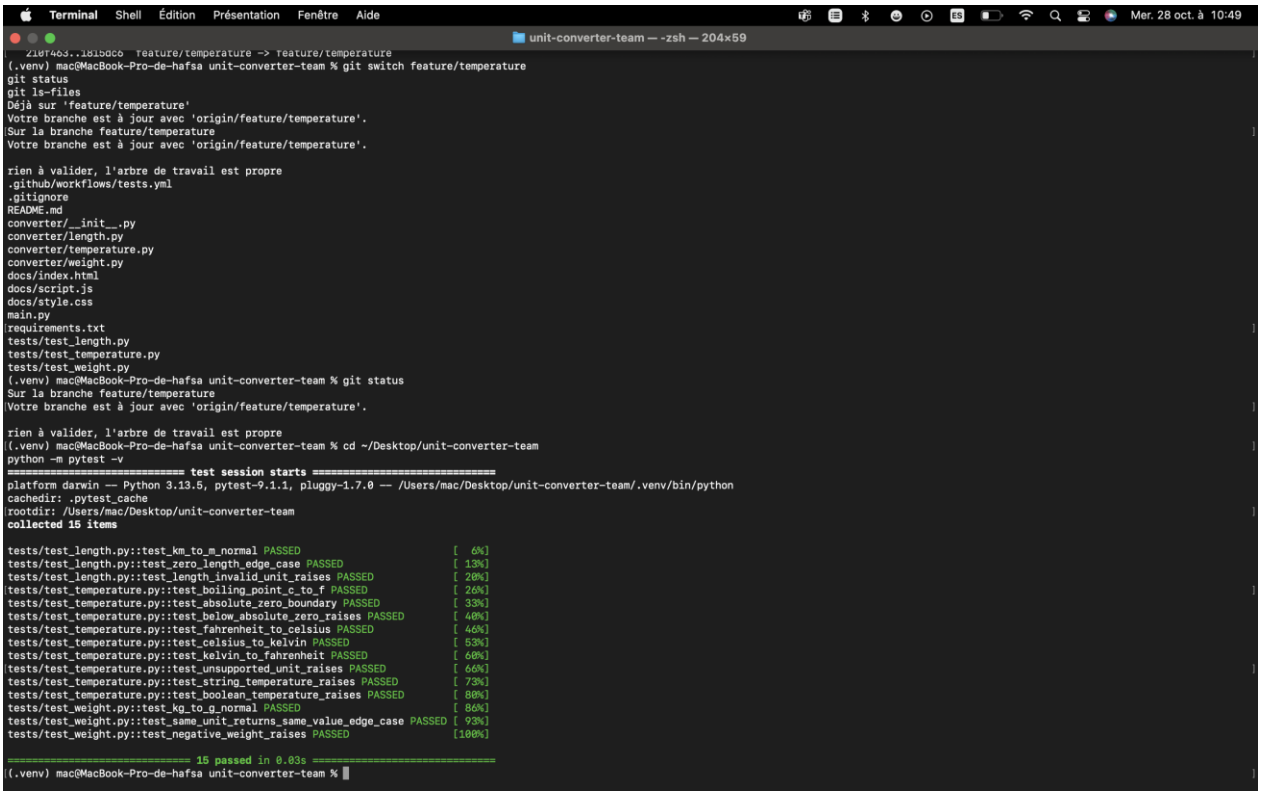
The tests cover normal conversions, boundary conditions, and invalid inputs to help verify that the conversion functions behave correctly.

Examples of test coverage

- **Length conversion:** Converting kilometers to meters, checking that zero length returns zero, and verifying that unsupported units or negative lengths raise an error.
- **Temperature conversion:** Converting Celsius to Fahrenheit and Kelvin, checking the absolute-zero boundary, and verifying that unsupported units and invalid input types are rejected.
- **Weight conversion:** Testing conversions between kilograms, grams, pounds, and ounces, along with invalid unit handling.

Test execution

The test suite was run locally using python -m pytest -v.



6. Continuous Integration

The workflow .github/workflows/tests.yml, named **Python Tests**, automates testing whenever code is pushed or a Pull Request is opened or updated. It uses an Ubuntu runner, sets up Python, installs the project dependencies, and runs the test suite with pytest.

- **Failed run:** [Insert link to the failed GitHub Actions run]
- **Cause:** [Describe the actual error shown in the failed run's logs.]

- **Fix:** [Describe the change made and include the commit link.]
- **Verification:** A later GitHub Actions run completed successfully, confirming that the tests passed after the fix.

7. Ruleset and Protected Main

The repository's ruleset is intended to protect the main branch by requiring changes to be submitted through Pull Requests and reviewed before merging. It also restricts branch deletion and blocks force pushes.

After the CI workflow was configured, the **Python Tests** status check was added as a required check, so Pull Requests must receive the required approval and pass the check before they can be merged.

The bypass list was not used.

Evidence: Screenshot of the active ruleset showing the enabled protections and required status checks, plus a screenshot of a Pull Request blocked from merging because a requirement was not satisfied.

8. Continuous Deployment

GitHub Pages is configured to publish the website from the main branch and the /docs folder. After a change was submitted through a Pull Request and merged into main, GitHub Pages deployed the updated website automatically.

Live website: https://zulaikhahumayun21.github.io/unit-converter/

9. Contribution Summary

Zulaikha Humayun

- **Feature:** Length conversion
- **Branch:** length
- **Pull Request:** Authored PR #2
- **Code review:** Reviewed PR #1 (Temperature conversion)
- **Tests:** Contributed to length conversion tests in tests/test_length.py.

Hafsa Borouja

- **Feature:** Temperature conversion
- **Branch:** feature/temperature
- **Pull Request:** Authored PR #1
- **Code review:** Reviewed PR #2 (Length conversion)

- **Tests:** Contributed to temperature conversion tests in tests/test_temperature.py.

10. Fair Use of AI

The group used AI assistance to clarify Git/GitHub commands, troubleshoot a rejected push caused by token permissions, draft review comments and organize report text. We adapted suggestions to our own code and branch names, ran the tests ourselves and verified results on GitHub before using them. Each member remains responsible for understanding the work attributed to their account.

11. Sources & Reflection

Sources: course Session 2 materials

https://github.com/zulaikhahumayun21/unit-converter.

The hardest problem was the merging part. If a PR, merge, test or workflow behaves unexpectedly, we would first check the Actions log, then the ruleset requirements, then whether the branch is up to date with main.